Product Manual | ICX-FieldHawk Rugged

Real-Time Spectrum Analyzer

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OVERVIEW

The ICX-FieldHawk Rugged series is a ruggedized, IP68-rated handheld real-time spectrum analyzer built for water, dust, and vibration exposure in field conditions, with certification testing aligned to MIL-STD-810H (method 512.6 for immersion, 516.8 for shock/drop, and 514.8 for vibration).

The instrument pairs a 10.1-inch full-touch display with a gesture-driven interface similar to a smartphone, in a package that weighs roughly 2.5 kg — light enough for sustained handheld use in the lab or field.

RF coverage spans 9 kHz up to 9.5, 20, or 40 GHz depending on model, with 100 MHz of standard analysis bandwidth. A fast-FFT design pushes sweep speed up to 1.0 THz/s.

An optional embedded AI compute module (NVIDIA Jetson Orin NX, 8 GB) adds up to 117 TOPS of on-device AI processing — an NVIDIA Ampere-architecture GPU with 32 Tensor Cores and 1024 CUDA cores at up to 1173 MHz — for fast, AI-assisted spectrum recognition and analysis directly on the instrument.

All models share SpecICX-gen3's common API across C/C++, C#, Python, MATLAB, Qt, and LabVIEW, simplifying secondary development and integration into existing test setups.

Key Features

- Rugged design: IEC 60529 IP68 ingress protection rating
- Environmental testing aligned to MIL-STD-810H, methods 512.6, 516.8, and 514.8
- Portable design: approx. 2.5 kg with a 10.1-inch multi-touch screen
- Battery life: approx. 4 hours typical
- Frequency range: 9 kHz to 9.5 / 20 / 40 GHz, depending on model
- Analysis bandwidth: 100 MHz
- 1 GHz DANL: < −160 dBm/Hz
- 1 GHz phase noise: < −100 dBc/Hz at 10 kHz offset
- Optional embedded AI compute module (NVIDIA Jetson Orin NX)
- Standard SCPI command support

SPECICX-GEN3 CONTROL SOFTWARE

SpecICX-gen3 is the companion control application for the ICX-FieldHawk Rugged series, offering seven primary operating modes: Standard Spectrum Analysis, IQ Streaming, Power Detection Analysis, Real-Time Spectrum Analysis, Digital Demodulation (Option 71), Harmonic Analysis, and Phase Noise Measurement.

Standard Spectrum Analysis Mode

Provides a full-span sweep together with common measurements such as channel power, OBW, ACPR, IM3, and spectral emission mask (SEM). Recording and playback of captured spectra are supported, and auxiliary tools — signal tracking, a peak table, and amplitude correction — combine into a single workspace for routine spectrum checks.

IQ Streaming Mode

Captures IQ data at up to 100 MHz of analysis bandwidth using several trigger methods. The mode displays IQ time-domain waveforms alongside spectrum and spectrogram views, and includes AM/FM demodulation and digital down-conversion (DDC).

Power Detection Analysis Mode

Analyzes time-domain signal behavior within the active analysis bandwidth, which suits applications centered on in-band power-versus-time relationships, such as characterizing pulsed signals.

Real-Time Spectrum Analysis Mode

Runs on a high-speed, FPGA-based FFT engine with gapless, non-overlapping frame processing, giving continuous real-time visibility across the full analysis bandwidth.

Digital Demodulation Mode (Option 71)

Demodulates and analyzes 2ASK, 2FSK, 4FSK, GMSK, BPSK, QPSK, 8PSK, and 16/64/128/256QAM signals, reporting constellation and modulation-quality measurements.

Harmonic Analysis Mode

Detects and measures up to 10 harmonic components, reporting harmonic peak levels, per-harmonic channel power, and total harmonic distortion.

Phase Noise Measurement Mode

Covers offset ranges from 1 Hz to 10 MHz for carrier phase-stability evaluation, with an automatic carrier-search routine that locates the target carrier without manual tuning.

MEASUREMENT FUNCTIONS

Beyond the primary operating modes, SpecICX-gen3 includes a standard toolset of measurement and workflow aids common across every mode:

- Channel Power — total power within a defined channel bandwidth
- OBW — occupied bandwidth for a percentage of total power

- ACPR — adjacent channel power ratio
- IM3 — third-order intermodulation measurement
- SEM — spectral emission mask pass/fail evaluation
- AM/FM Demodulation — modulation depth, deviation, and audio-domain analysis
- Data Record and Playback — capture and replay of spectrum data
- Pulse Detection (Option 72) — pulse timing and level parameter measurement
- Antenna Factor — correction curve applied for calibrated field-strength measurements
- Amplitude Offset — external attenuation/gain compensation
- Signal Track — automatic peak tracking across sweeps
- Peak Table — sortable list of detected signal peaks
- Multiple Unit Display — simultaneous multi-unit view for distributed monitoring
- Orin NX Super (Option 08) — on-device AI compute for accelerated spectrum recognition

SPECIFICATIONS

Frequency

Parameter	ICX-095R	ICX-200R	ICX-400R
Frequency range	9 kHz to 9.5 GHz	9 kHz to 20 GHz	9 kHz to 40 GHz
Reference clock	Internal or external		
Frequency accuracy — TCXO (std.)	< 1 ppm, manual correction available		
Frequency accuracy — OCXO (Option 01)	< 1 ppm, manual correction available		
Aging & temperature stability — TCXO (std.)	< 1 ppm/year, < 1 ppm		
Aging & temperature stability — OCXO (Option 01)	< 1 ppm/year, < 0.15 ppm		

Spectrum Purity

SSB Phase Noise (dBc/Hz)

Carrier frequency	ICX-095R		ICX-200R		ICX-400R	
	1 GHz	9.5 GHz	1 GHz	20 GHz	1 GHz	40 GHz
1 kHz	-95.2	-91.5	-91.2	-80.6	-99.0	-78.4
10 kHz	-101.6	-98.5	-99.7	-90.6	-107.5	-85.7
100 kHz	-100.6	-99.7	-101.1	-96.2	-107.7	-85.1
1 MHz	-120.9	-116.2	-121.6	-111.5	-122.7	-100.8

Residual Response (dBm) — Spur Reject = Bypass, RBW = 1 kHz, PosPeak Detector

Frequency range	ICX-095R		ICX-200R		ICX-400R	
	0 dBm	-50 dBm	0 dBm	-50 dBm	0 dBm	-50 dBm
9 kHz to 1 GHz	-83	-120	-90	-120	-72	-103
1 GHz to 3 GHz	-83	-120	-80	-120	-72	-103
3 GHz to 10 GHz	-90	-130	-90	-120	-72	-103
10 GHz to 20 GHz	-	-	-90	-120	-91	-115
20 GHz to 40 GHz	-	-	-	-	-85	-105

IF Rejection (dBc), Typical

Model	Spur reject: enhanced	Spur reject: bypass
ICX-095R	> 90 dBc	> 80 dBc
ICX-200R	> 90 dBc	> 80 dBc
ICX-400R	8.2–21.75 GHz: > 68 dBc; other bands: > 90 dBc	—

Image Rejection (dBc), Typical

Frequency range	ICX-095R		ICX-200R		ICX-400R	
	standard	bypass	standard	bypass	standard	bypass
90 MHz to 3 GHz	> 90	> 76	> 90	> 79	> 90	—
3 GHz to 9.5 GHz	> 90	> 60	> 90	> 68	> 90	—
9.5 GHz to 20 GHz	-	-	> 90	> 60	> 90	—
20 GHz to 33 GHz	-	-	-	-	> 90	—
33 GHz to 40 GHz	-	-	-	-	> 58	—

Parameter	Specification
Local-oscillator-related spurious	< -65 dBc, center frequency ± (N/M) × 125 MHz, N,M = 1,2,3,4,5...

IIP3 / IIP2 (dBm), Typical

Carrier frequency	ICX-095R		ICX-200R		ICX-400R	
	1 GHz	9.5 GHz	1 GHz	20 GHz	1 GHz	40 GHz
R.L. = 20 dBm	46.1 / 83.2	40.5 / 92.8	45.5 / 82.6	35.3 / 93.6	40.3 / 75.5	31.7 / 88.6
R.L. = 0 dBm	26.7 / 85.0	19.2 / 90.3	25.5 / 81.1	21.0 / 89.0	27.4 / 45.3	10.3 / 86.1
R.L. = -20 dBm	10.5 / 82.2	2.0 / 49.3	7.9 / 81.5	-4.5 / 55.3	8.7 / 25.2	4.8 / 66.6

Amplitude

Parameter	Specification
Display range	DANL to 23 dBm typ. (ICX-095R / ICX-200R); DANL to 20 dBm typ. (ICX-400R)
Reference level	−50 dBm to +23 dBm typ. (ICX-095R / ICX-200R); −50 dBm to +20 dBm typ. (ICX-400R)
VSWR	90 MHz to 9.5/20 GHz: < 2.0:1 (ICX-095R / ICX-200R); 90 MHz–16 GHz: < 2.0:1, 16–40 GHz: < 3.0:1 (ICX-400R)
Max. DC voltage	± 10 VDC
IF in-band flatness	± 2.0 dB
Max. input power (CW)	23 dBm (50 MHz to max. frequency, preamplifier off); 10 dBm (9 kHz to 50 MHz, or preamplifier on)
Amplitude accuracy	9 kHz to 9.5 GHz: ± 2.0 dB; 9.5 GHz to 20/40 GHz: ± 3.0 dB
RF preamplifiers	Automatic on, or forced off

Displayed Average Noise Level (DANL), dBm/Hz, RBW = 1 kHz

Frequency range	ICX-095R		ICX-200R		ICX-400R	
	-20 dBm	-50 dBm	-20 dBm	-50 dBm	-20 dBm	-50 dBm
9 kHz to 1 MHz	-143.0	-152.4	-143.6	-152.6	-136.0	-145.8
1 MHz to 90 MHz	-152.0	-159.2	-151.8	-160.0	-153.7	-158.0
90 MHz to 3 GHz	-146.0	-167.5	-149.7	-166.3	-154.1	-159.9
3 GHz to 9.5 GHz	-153.6	-167.0	-151.4	-157.5	-154.1	-159.9
9.5 GHz to 19 GHz	-	-	-156.1	-160.6	-156.8	-161.5
19 GHz to 20 GHz	-	-	-156.1	-160.6	-145.2	-149.3
20 GHz to 40 GHz	-	-	-	-	-145.2	-149.3

Standard Spectrum Analysis

Parameter	Specification
Detector	PosPeak, NegPeak, Sample, Average, RMS, MaxPower
RBW	1 Hz to 10 MHz
VBW	1 Hz to 10 MHz
Data chart	SpecICX-gen3 provides spectrum, spectrogram, and historical-trace views
Measurements	Channel power, OBW, X dB bandwidth, adjacent channel power ratio, IM3
Sweep speed (RBW = 250 kHz, FPGA, spur reject = bypass)	1.0 THz/s

Parameter	Specification
Sweep speed (RBW = 250 kHz, FPGA, spur reject = standard)	558.8 GHz/s
Sweep speed (RBW = 50 kHz, FPGA, spur reject = bypass)	212.6 GHz/s
Sweep speed (RBW = 1 kHz, CPU, spur reject = bypass)	2.6 GHz/s

Detection Analysis

Parameter	Specification
Lowest time resolution	8 ns
Max. analysis bandwidth	100 MHz
Detector	PosPeak, NegPeak, Sample, Average, RMS, MaxPower

IQ Recording

Parameter	Specification
Burst recording bandwidth	Maximum 100 MHz; built-in memory depth 128 Mbytes
Continuous recording bandwidth	Maximum 25 MHz
IQ sample rate	Maximum 125 MSPS; decimate factor 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, 2048, 4096
External trigger response	Maximum 500 times/s

Real-Time Spectrum Analysis

Parameter	Specification
FFT analysis	FPGA-implemented FFT engine with frame compression and trace detection; no missing samples between FFT frames. FFT frame update rate = 10^9 ns / (N × D × 8 ns); POI = 2 × N × D × 8 ns, where N is FFT points (32 to 2048) and D is decimate factor (1, 2, 4, 8...)
Max. analysis bandwidth	100 MHz
Window function	B-Nuttall, Flat-top, LowSideLobe
RBW	14.73 MHz to 3.59 kHz (Flat-top); 7.81 MHz to 1.90 kHz (B-Nuttall); 13 grades per window type
Amplitude resolution	0.75 dB

Typical FFT Settings

Typical setting	FFT refresh rate	POI
N = 2048, D = 1	61,035 times/sec	32.768 µs
N = 32, D = 1	3,906,250 times/sec	0.512 µs

Environmental Adaptability

Parameter	Specification
Water and dust resistance	IEC 60529 IP68 rating; testing aligned to MIL-STD-810H, method 512.6
Drop resistance	Testing aligned to MIL-STD-810H, method 516.8
Vibration resistance	Testing aligned to MIL-STD-810H, method 514.8

General

Parameter	Specification
RF input	N-type female, 50 Ω (ICX-095R / ICX-200R); 2.4 mm male, 50 Ω (ICX-400R)
Power	USB Power Delivery, 65 W
USB ports	USB 3.0 Type-C ×1, USB 2.0 Type-C ×1, USB 2.0 Type-A ×1
Audio / display interface	Micro HDMI ×1 (extended display support); 3.5 mm headphone port ×1
External reference clock input	MMCX female, 10 MHz, amplitude ≥ 1.5 Vpp, ≈ 330 Ω impedance
Reference clock output	Integrated in Aux I/O, 10 MHz, 3.3 V CMOS, programmable on/off
External trigger input	MMCX female, 3.3 V CMOS, high impedance
Trigger output	MMCX female, 3.3 V CMOS
External antenna input	MMCX female
Analog IF output	MMCX female, max. output power −25 dBm; 50 Ω impedance, 307.2 MHz ± 50 MHz
Display	IPS LCD, 1280 × 800, 10.1-inch multi-touch
RAM / eMMC storage	4 GB / 32 GB
Power consumption	25 W typ. (standard configuration)
Battery life	Approx. 4 hours typ.; external power bank supported
Dimensions (D × W × H)	285 × 208 × 58 mm
Weight	2.5 kg
GNSS 1PPS synchronization accuracy	± 100 ns; built-in GNSS, external antenna only
Operating temperature (ambient)	T1 class (std.): −20 to +65°C

Parameter	Specification
Storage temperature (ambient)	T1 class (std.): −40 to +85°C
Packaging and accessories	Protected main unit ×1, power adapter ×1, power cord ×1, lanyard ×1

Specifications apply under the following conditions: (1) instrument powered on and warmed up for 10 minutes; (2) ambient temperature 25°C, core temperature 50°C; (3) Standard Spectrum Analysis mode, spurious rejection set to standard; (4) adequate heat dissipation is provided to keep ambient and core temperature within the rated range; (5) sweep-speed and DANL figures were measured against a specific internal firmware baseline — pending confirmation of the equivalent SpecICX-gen3 build — and typical/nominal values are not guaranteed and exclude measurement uncertainty; specifications may change with hardware and software revisions without prior notice.

OPTIONS

Code	Description	Type
01	Built-in OCXO reference clock	Built-in hardware
08	Built-in Orin NX Super processing platform	Built-in hardware
34	External omnidirectional antenna, 400–8000 MHz, gain < 2 dBi	Accessory
35	External active directional antenna, 0.5–10 GHz; gain 5 dBi (amp off), 25 dBi (amp on)	Accessory
71	Basic digital demodulation	Software
72	Pulse detection	Software

Orin NX Super (Option 08)

Parameter	Specification
AI performance	117 TOPS
GPU	NVIDIA Ampere-architecture GPU, 32 Tensor Cores, 1024 CUDA cores
GPU max. frequency	1173 MHz
RAM	8 GB
Built-in SSD storage	256 GB std.; 512 GB / 1 TB optional
Software compatibility	Compatible with NVIDIA JetPack SDK, to accelerate development and simplify deployment